

Initial Business Insights

A. Five Workforce Observations

1. The company has a stable workforce.

Approximately 83.9% of employees remain with the company, while 16.1% have left. This indicates that most employees stay, although attrition is still a concern.

2. Research & Development is the largest department.

The Research & Development department has the highest number of employees, followed by the Sales department. Human Resources has the smallest workforce.

3. Most employees do not work overtime.

The majority of employees reported not working overtime, suggesting that overtime is limited to a smaller group of the workforce.

4. The workforce is relatively young and experienced.

The average employee is about 37 years old, with an average of 11 years of total working experience and approximately 7 years at the company.

5. Most employees travel only occasionally.

The majority of employees fall under the Travel_Rarely category, indicating that frequent business travel is uncommon.

B. Five Possible Factors Contributing to Attrition

1. Overtime may increase employee attrition.

Employees who work overtime appear more likely to leave the company than employees who do not work overtime. This suggests that long working hours may contribute to stress and lower job satisfaction.

2. Younger employees are more likely to leave.

The boxplots show that employees who left the company are generally younger than those who stayed. Younger employees may be more likely to seek career growth or better opportunities elsewhere.

3. Lower monthly income is associated with higher attrition.

Employees who left tend to earn lower monthly salaries than employees who remained. This suggests that compensation may influence an employee's decision to stay.

4. Employees with shorter tenure are more likely to leave.

Employees with fewer YearsAtCompany, YearsInCurrentRole, and TotalWorkingYears show higher attrition. This indicates that newer and less experienced employees are more likely to resign.

5. Certain departments and job roles experience higher turnover.

The Research & Development and Sales departments, particularly roles such as Laboratory Technician, Sales Executive, and Research Scientist, show relatively higher numbers of employees leaving. These positions may require additional attention to improve employee retention.

Conclusion

The exploratory analysis suggests that overtime, age, salary, work experience, length of service, and job role are the variables most closely associated with employee attrition. These factors should be investigated further and considered when developing future predictive machine learning models to identify employees who may be at risk of leaving.